Name: ______________________

PERCENTAGES OF AMOUNTS, (FINDING ANY PERCENTAGE OF AN AMOUNT)

Date: ________________

LO: I can find any percentage of an amount using a calculator or written method.

Finding any percentage

To find any percentage of an amount, write the percentage as a decimal or fraction and multiply. This method works for whole and decimal percentages.

Method: Convert the percentage to a decimal (divide by 100), then multiply by the amount.

Quick examples

●	23% of 400 = 0.23 × 400 = 92	decimal multiplier
●	7.5% of 80 = 0.075 × 80 = 6	decimal percentage
●	110% of 50 = 1.10 × 50 = 55	more than 100%
●	23% of 400 = 23 × 400	must divide 23 by 100 first

Key idea

Any percentage can be found by converting to a decimal and multiplying.

$$P\% \text{ of } n = {}^{P}\!/_{100} \times n$$

Works for any P, including decimals and values over 100

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – whole number percentage

Find 37% of 250

Convert: 37% = 0.37

0.37 × 250

= 92.5

= 92.5

Divide the percentage by 100 to get the decimal multiplier.

Example 2 – decimal percentage

Find 3.5% of 600

Convert: 3.5% = 0.035

0.035 × 600

= 21

= 21

Decimal percentages work the same way.

Example 3 – percentage over 100

Find 125% of 80

Convert: 125% = 1.25

1.25 × 80

= 100

= 100

Percentages over 100% give values larger than the original.

Example 4 – real-world context

A phone costs £480. There is 3.5% cashback. How much cashback?

Convert: 3.5% = 0.035

0.035 × 480

= £16.80

= £16.80

Same decimal multiplier method applied to money.

YOUR TURN – PRACTICE (deliberate practice)

1. Whole number percentages

Convert to a decimal and multiply.

a) 23% of 300

b) 47% of 200

c) 68% of 50

d) 89% of 400

2. Decimal percentages

Find each percentage.

a) 2.5% of 800

b) 4.5% of 200

c) 0.5% of 1000

d) 12.5% of 64

3. Percentages over 100

Calculate percentages greater than 100%.

a) 150% of 60

b) 200% of 45

c) 175% of 80

d) 110% of 250

4. Real-world percentages

Solve each problem.

a) A meal costs £65. Service charge is 12.5%. How much is the charge?

b) A savings account has £3400. Interest rate is 1.5%. How much interest?

c) 33% of 270 students walk to school. How many?

d) A laptop was £800. It has lost 17.5% of value. Loss in pounds?

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick ( ) the ones that are correct. If it is wrong, write the correct answer.

a) 37% of 250 = 0.37 × 250 ☐

Correct answer: __________

Reason: _________________

b) 7.5% = 0.75 as a decimal ☐

Correct answer: __________

Reason: _________________

c) 125% of 80 is greater than 80 ☐

Correct answer: __________

Reason: _________________

d) 0.5% of 1000 = 50 ☐

Correct answer: __________

Reason: _________________

e) 3.5% = 0.035 as a decimal ☐

Correct answer: __________

Reason: _________________

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A company increases salaries by **3.2%**. An employee earns **£28500**. What is their new salary? Another employee wants to earn at least **£30000** after the rise. What minimum current salary do they need?

Expression: ____________________________

Simplified: ____________________________

b) In a survey of **850 people**, **37.6%** prefer tea, **28.8%** prefer coffee, and the rest prefer water. How many people prefer each drink? Round to the nearest whole number.

Expression: ____________________________

Simplified: ____________________________

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

WELL DONE!